

Rohan Batra

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Professional Summary

Computer Science undergraduate (UPES, '27) focused on reliability and platform engineering — building and operating production systems on Linux and cloud infrastructure, and root-causing incidents end-to-end. I design and operate UPES-ECS, an always-on emergency-communication system on enterprise networking (active/standby HA, health checks, load testing, runbooks). HACKIITK 2026 finalist at IIT Kanpur (top ~0.4% of 1,300+ teams). Comfortable across Linux, AWS, Docker, CI/CD, and security audits (OWASP, SAST/DAST) throughout the SDLC.

Education

B.Tech in Computer Science Engineering, UPES Dehradun Aug 2027
CGPA: 8/10

Technical Skills

Reliability & Systems: Linux, systemd, High availability / failover, Health checks, Incident response, Load testing, Runbooks, Proxmox
Cloud, DevOps & Automation: AWS (EC2, ECS, S3), Google Cloud, Docker, GitHub Actions, CI/CD, System Automation
Networking: Juniper (SRX, EX, Mist), TCP/IP, DNS, VLAN, QoS
Software Engineering: Python, C/C++, JavaScript, SQL, FastAPI, REST APIs, Git
Security & Accessibility: OAuth2, RBAC, OWASP Top 10, SAST/DAST, WCAG 2.0
AI & Tooling: LangChain, Pydantic, Ollama, Redis
Web & Mobile: Hugo, Flutter, Android SDK, HTML, CSS

Experience

Techmonks, Bangalore (Remote) Nov 2025 – Present
Software Engineer Intern

- Building Python/FastAPI backend services deployed on AWS (EC2, ECS, S3)
- Conducting application security audits combining manual code review, automated scanning (SAST/DAST), and vulnerability reproduction against OWASP Top 10
- Integrating automated security checks into CI/CD pipelines as part of the continuous integration workflow
- Contributing to WCAG 2.0 accessibility on frontend integrations and backend-supported features

Google Developer Groups On Campus (GDGOC), UPES Dehradun Sep 2025 – Present
Technical Head

- Led and delivered technical workshops for 100+ participants
- Managed cross-functional teams executing production-grade initiatives
- Mentored students in cloud, system design, and full-stack development

LinuxWale, Dehradun Sep 2023 – Apr 2025
Linux Support & Community Volunteer

- Onboarded 10+ students to Linux: distribution selection, dual-boot setup, and driver/hardware troubleshooting across varied laptops
- Wrote shell scripts to automate post-install configuration and routine setup tasks
- Mentored peers transitioning from Windows to Linux as a daily-driver operating system

Cybersecurity Awareness Sessions, Indian Armed Forces Cantonment Welfare Programs Aug 2024 – Sep 2024
Cybersecurity Awareness Trainer

- Delivered cybersecurity and social media safety awareness sessions within Indian Armed Forces cantonment premises across three official welfare programs
- Engaged with families of Indian Armed Forces personnel and senior officers through practical sessions on digital safety, privacy risks, online fraud prevention, and responsible social media usage
- Adapted technical cybersecurity concepts for diverse audiences, including spouses, children, and non-technical participants
- Recognized with three Certificates of Appreciation from Golden Key Eagles Women Empowerment Center, HQ Uttarakhand Sub Area, and Golden Key Tuskers

Student Volunteer

- Supported sports programs for visually impaired participants
- Coordinated logistics and on-ground event execution
- Contributed to accessibility-focused initiatives

Projects

UPES-ECS — Campus Emergency Communication System

2026

Showcased to Team HPE | rohanbatrain.github.io/UPES-ECS

Tech Stack: Linux, Asterisk/PJSIP (VoIP/SIP), Python/FastAPI, Docker, HPE Juniper (SRX/EX/Mist)

- Designed and operate a resilient, offline-first campus emergency communication system — one number (111) reaches a trained response team with no internet, cloud, or cellular dependency
- Runs on an enterprise HPE + Juniper networking backbone (SRX firewall, EX2300 PoE switches, Mist Wi-Fi 6) with VLAN/QoS prioritization and high-availability failover
- Built a Python/FastAPI operations console and telemetry API with an automated call flow, offline voice coaching, and 44-language localized prompts generated offline
- Presented live to Team HPE during their campus visit; the architecture extends beyond campus to disaster-response and critical-communication zones

HACKIITK Finale — Gap Hunter (Team Hunterz)

2026

Organized by C3iHub, IIT Kanpur

Tech Stack: Python, LangChain, Ollama, Pydantic, Electron, GitHub Actions

- Built an offline AI-powered policy gap analyzer mapping security policies against NIST CSF 2.0 and the CIS MS-ISAC framework, producing evidence-grounded gap reports, revised policy drafts, and remediation roadmaps
- Designed a multi-stage pipeline (PDF/DOCX ingestion → hybrid retrieval combining vector + sparse + reranking → gap analysis → policy revision) orchestrated via LangChain with Pydantic-validated structured outputs to ground LLM reasoning and reduce hallucination
- Shipped cross-platform Electron desktop app with local Ollama runtime integration; automated Linux/macOS/Windows releases via GitHub Actions and PyInstaller
- Maintained 702-test suite (97.7% pass rate) spanning unit, integration, property-based, and adversarial/chaos tests
- Placed top ~0.4% of 1,300+ teams and 9,000+ participants in the Solutions Track; selected as one of 24 finalist teams invited to IIT Kanpur campus

Second Brain Database

Nov 2024

Tech Stack: Python, FastAPI, Redis, OAuth2, RBAC, Docker

- Built secure REST APIs using OAuth2 authentication and role-based access control (RBAC)
- Designed optimized database schemas and integrated Redis for caching
- Containerized services using Docker for consistent CI/CD deployments

n8n Nodes — Second Brain Database

Jan 2025

Tech Stack: JavaScript, n8n, Workflow Automation

- Developed custom n8n nodes exposing Second Brain Database APIs as reusable workflow steps
- Enabled end-to-end automation pipelines integrating backend services with orchestration workflows

Portfolio & Resume CI/CD Automation

Jan 2025

Tech Stack: Hugo, Flutter, MarkLang, GitHub Actions

- Built multi-source deployment pipeline merging a Hugo blog and a Flutter portfolio into a unified gh-pages deployment
- Integrated MarkLang as Hugo's content translation layer for blog posts, bridging a custom language with a static site generator
- Automated build, merge, and deploy on commit; eliminated manual release steps for both portfolio and resume

MarkLang (Translation Engine)

Jan 2025

- Built an Ollama-powered translation engine for automated multilingual workflows
- Automated the full translation pipeline using GitHub Actions for autonomous execution

Proxmox Auto-Install Assistant

Jan 2025

- Automated installation of Proxmox, reducing setup time by ~80%
- Simplified complex system configuration workflows through scripted provisioning

Karmstrot Builds (Android & C++)

Aug 2024

Tech Stack: C++, Shell, Android SDK, KernelSU

- Built automation scripts for Android KernelSU integration across devices
- Modified low-level system configurations and applied SDLC practices for versioning and testing

Certifications & Accomplishments

- Top 15% — TryHackMe cybersecurity learning platform
- Certificate of Appreciation — Golden Key Eagles Women Empowerment Center (Sep 2024)
- Certificate of Appreciation — HQ Uttarakhand Sub Area (Aug 2024)
- Certificate of Appreciation — Golden Key Tuskers, AWWA Week 2024 (Aug 2024)
- Semifinalist — Eureka! Junior 2022–23, E-Cell IIT Bombay (Issued Nov 2022; Credential ID: ecell2021)
- Semifinalist — Eureka! Junior 2021–22, E-Cell IIT Bombay (Issued Dec 2021; Credential ID: ecell2021)